Chapter 5 Large-scale Stochastic Impulsive Systems with Time Delay.
Details (refer to "Stability and stabilization of large-scale stochastic impulsive systems with time delay" 2019. Dynam. ~~&~~ Continuous Discrete Impuls. Syst ~~em~~ B Appl. Algorithm.)
Futher work : Input-to-state stability for large-scale stochastic impulsive systems with state delay. Stochastic Analysis and Applications. 2023.

5.1 Problem Formulation

$\boxed{\text{Interconnected}}$ system with decomposition $D_i$ $(i=1,\cdots,l)$ :

$$D_i:\ (5.1)\begin{cases} dW^i(t) = f_i(t, W_t^i)\,dt + \underbrace{g_i(t, W_t^1, W_t^2,\cdots, W_t^l)}_{\text{coupled}}dt + \underbrace{\sum_{\substack{j=1\\ i\neq j}}^{l} \sigma_{ij}(t, W_t^i)\, dW_j(t)}_{\text{coupled}}, & t\neq \tau_k \\ \Delta W^i(\tau_k) = \mathcal{I}_i(\tau_k, W^i_{\tau_k^-}), \\ W^i_{t_0} = \phi_i(s), \quad s\in[-r,0]. \end{cases}$$

where $W^i\in\mathbb{R}^{n_i}$, $f_i:\mathbb{R}_+\times\mathbb{R}^{n_i}\to\mathbb{R}^{n_i}$, $g_i:\mathbb{R}_+\times\mathbb{R}^{n}\to\mathbb{R}^{n_i}$ $\quad (n=\sum_{i=1}^{l} n_i)$.
$\sigma_{ij}:\mathbb{R}_+\times\mathbb{R}^{n_j}\to\mathbb{R}^{n_i\times m_j}$. $W_j$ — $m_j$-demensional Wiener process.
$\phi_i:[-r,0]\to\mathbb{R}^{n_i}$, $\mathcal{I}_i: T\times\mathbb{R}^{n_i}\to\mathbb{R}^{n_i}$, $T\triangleq\{\tau_k|k\in\mathbb{N}\}$ — impulsive moments.
$0<\tau_1<\tau_2<\cdots$ and $\lim_{k\to\infty}\tau_k=\infty$. $\quad (m=\sum_{j=1}^{n} m_j)$

ignoring the interconnection between the subsystems

Isolated subsystem $S_i$ :

$$S_i:\ (5.2)\begin{cases} dW^i(t) = f_i(t, W_t^i)\,dt + \sigma_{ii}(t, W_t^i)\, dW_i(t), & t\neq\tau_k \\ \Delta W^i(\tau_k) = \mathcal{I}_i(\tau_k, W^i_{\tau_k^-}) \\ W^i_{t_0} = \phi_i(s). \quad s\in[-r,0]. \end{cases}$$

$\boxed{\text{Goal}}$: Find Lyapunov-like sufficient conditions under which (5.1) has uniform asymptotic stability in the m.s.
$\boxed{\text{Technique}}$: Lyapunove-Razumikhin technique. comparision method.

Process : Step A: Focus on the isolated subsystem $S_i$, $\xrightarrow{\text{Results in Chapter 4}}$ $\boxed{\text{Property A}}$
uniform asymptotic stability of $S_i$ $\boxed{\text{Property B}}$

Step B : Consider the interconnected part as perturbation.
Find suitable conditions to guarantee uniform stability of $D_i$ $(i=1,\cdots,l)$ ⇐ Results in Chapter. 4.

Notations :

$x^T = [(W^1)^T, (W^2)^T, \cdots (W^L)^T] \in \mathbb{R}^n$ —— state

vector field $f : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^n$ with $f^T(t, x_t) = [f_1^T(t, W_t^1), \cdots, f_L^T(t, W_t^L)]$

interconnection $g : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^n$ with $g^T(t, x_t) = [g_1^T(t, x_t), \cdots, g_L^T(t, x_t)]$.

diffusion matrix function $\sigma : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^{n \times m}$ with $\sigma(t, x_t) = [\sigma_{ij}(t, W_t^j)]$

Wiener process vector $W : \mathbb{R}_+ \to \mathbb{R}^m$ with $W^T = [W_1^T, \cdots, W_L^T]$.

impulsive functional vector $J : T \times \mathbb{R}^n \to \mathbb{R}^n$ with

$J^T(t, x_{t^-}) = [J_1^T(t, W_{t^-}^1), \cdots, J_L^T(t, W_{t^-}^L)]$.

initial vector state $\Phi : [-r, 0] \to \mathbb{R}^n$ with $\Phi^T = [\phi_1^T, \cdots, \phi_L^T]$.

Rewrite (5.1) as follows

$$(5.3) \begin{cases} dx(t) = F(t, x_t)\, dt + \sigma(t, x_t)\, dW(t), & t \neq \tau_k \\ \Delta x(\tau_k) = J(\tau_k, x_{\tau_k^-}) \\ x_{t_0} = \Phi(s), & s \in [-r, 0]. \end{cases}$$

where $F(t, x_t) = f(t, x_t) + g(t, x_t)$. $F \in \mathcal{L}ad(\Omega; L[t_0, t_0+\alpha])$

$\sigma \in \mathcal{L}ad(\Omega; L^2[t_0, t_0+\alpha])$. $\Phi \in \mathcal{L}^2_{F_0}([-r, 0]; \mathbb{R}^n)$. $\alpha > 0$.

Definition of uniformly asymptotically stable in the m.s

$x \equiv 0$ of (5.3) is said to be uniformly asymptotically stable in the m.s. if

$\forall \varepsilon > 0$. $t_0 \in \mathbb{R}_+$. $\exists\ \delta = \delta(\varepsilon) > 0$. s.t.

$E[\|\phi\|_r^2] \leq \delta$ implies $E[\|x(t)\|^2] < \varepsilon$. $\forall t \geq t_0$.

where $x(t) = x(t; t_0, \phi)$ is a solution of (5.3). $x \in PC([t_0 - r, t_0+\alpha]; D)$.

and $\Phi \in \mathcal{L}^2_{F_0}([-r, 0]. D)$.

Remark :

(5.3) looks like the system in Chapter 4. But we can not use the results in Chapter 4 directly to develop Lyapunov-like conditions to ensure the stability property since the interconnection part $g$ and $\sigma_{ij}$ $(i \neq j)$. That is the reason we focus on $S_i$ first.

Recall results in Chapter 4.

$$\begin{cases} dx(t) = f(t, x_t)\,dt + g(t, x_t)\,dW(t), & t \neq \tau_k \\ \Delta x(t) = J(t, x_{t^-}) & t = \tau_k \\ x_{t_0} = \phi(s), & s \in [-r, 0] \end{cases}$$

Assumption A1. $\exists\, 0 \le \rho_1 \le \rho$ s.t. for all $\tau_k \in \mathbb{R}_+$, $x \in PC([-r,0]; D)$ for some $D \subseteq \mathbb{R}^n$ (open)

if $E[\|x(\tau_k^-)\|^2] < \rho_1$ then $E[\|x(\tau_k)\|^2] < \rho$.

Assumption A2 For any $k \in \mathbb{N}$,

$\tau_{sup} = \sup\{\tau_k - \tau_{k-1}\} < \infty$, $\tau_{inf} = \inf\{\tau_k - \tau_{k-1}\} > 0$.

Corollary 4.1

Assumption A1 and A2 hold, Let $V \in C^{1,2}([-r,\infty) \times \mathbb{R}^n; \mathbb{R}_+)$ s.t.

(i) $\exists\, a \in \mathcal{K}_C$, $b \in \mathcal{K}_V$, s.t. $\forall (t, \psi(0)) \in [-r, \infty) \times S(\rho)$,

$b(\|\psi(0)\|^2) \le V(t, \psi(0)) \le a(\|\psi(0)\|^2)$. (a.s.)

(ii) $\exists\, c \in \mathcal{K}_C$, $p \in PC(\mathbb{R}_+, \mathbb{R}_+)$, s.t. $\forall t \neq \tau_k$ and $\psi \in PC([-r,0]; S(\rho))$

$\mathcal{L} V(t, \psi(0)) \le -p(t)\, c(V(t, \psi(0)))$. (a.s.)

provided that $V(t+s, \psi(s)) \le \bar{g}(V(t, \psi(0)))$ for some $s \in [-r, 0]$.

where $\bar{g} \in \mathcal{K}_3$.

(iii) $\exists\, \hat{g} \in \mathcal{K}_3$, s.t. at any impulsive moment $\tau_k \in T$.

$\forall \psi \in PC([-r, \infty); S(\rho))$.

$V(\tau_k, \psi(0) + J(\tau_k, \psi(\tau_k^-))) \le \hat{g}(V(\tau_k, \psi(0)))$. (a.s.)

with $\psi(0^-) = \psi(0)$. where $(\tau_k, \psi(\tau_k^-)) \in \mathbb{R}_+ \times PC([-r,0]; S(\rho_1))$.

(iv) $M_2 = \sup_{q \ge 0} \int_q^{\bar{g}(q)} \frac{ds}{c(s)} < \infty$. and $M_1 = \inf_{t \ge 0} \int_t^{t+\mu} p(s)\,ds > M_2$

with $\mu = \inf\{\tau_k - \tau_{k-1}\} > 0$.

Then the trivial solution $x \equiv 0$ of (4.1) is uniformly asymptotically stable in the m.s.

For $S_i$ $(i = 1, \cdots, l)$ $\boxed{\text{Property A}}$ ← corollary 4.1

$V^i \in C^{1,2}([-r,\infty) \times S(\rho); \mathbb{R}_+)$. $a_i, c_i \in \mathcal{K}_C$ $b_i \in \mathcal{K}_V$. $p = -\sigma_i > 0$ constant.

$\bar{g}_i \in \mathcal{K}_3$. $\hat{g}_i \in \mathcal{K}_3$. $M_2^i < -\sigma_i \mu$.

$\mathcal{K}_c \triangleq \{ a \in C([0,\rho) ; \mathbb{R}_+) : a(0)=0 .\ \text{concave} .\ \nearrow_+ \}$.
$\mathcal{K}_v \triangleq \{ b \in C([0,\rho) ; \mathbb{R}_+) : b(0)=0 .\ \text{convex} .\ \nearrow_+ \}$.
$\mathcal{K}_3 \triangleq \{ g \in C(\mathbb{R}_+ . \mathbb{R}_+) : g(0)=0 .\ \text{concave}\ \nearrow \}$.
$PC([a,b] ; D) = \{ \psi : [a,b] \to D \mid$ discontinuous points are at most finite. $\psi(t^+)=\psi(t)$ (right-continuous). $\psi(t^-)$ exist $\}$

Theorem 5.1 $\leftarrow$ ~~Coll~~ Corollary 4.1
(5.3) satisfies
(i) $S_i$ possesses Property A.
(ii) $\exists\ b_{ij} > 0 .\ \bar{q} \geq 1 .$ s.t.
$$g_i^T(t, \psi) V_{\psi^i(0)}^i (t, \psi^i(0)) \leq C_i^{1/2}(\|\psi^i(0)\|^2) \sum_{j=1}^{L} \bar{q}\, b_{ij}\, C_j^{1/2}(\|\psi^j(0)\|^2).$$
(iii) $\exists\ e_i > 0 .$ s.t.
$$(y^i)^T V^i_{\psi^i(0)\psi^i(0)} (t, \psi^i(0))\, y^i \leq \bar{q}\, e_i \|y^i(0)\|^2 .$$
where $y^i = \sigma_{ij}(t, \psi^j)$. $\rightarrow C_i(\|\psi^i(0)\|^2)$
(iv) $\exists\ d_{ij} \geq 0 .$ s.t. $\|\sigma_{ij}(t, \psi^j)\|^2 \leq d_{ij}\, C_i(\|\psi^j(0)\|^2)$.
or. $C_i = C_j\ \forall i \neq j$

(v). $S = [s_{ij}]_{i \times j}$ is negative definite where
$$s_{ij} = \begin{cases} \alpha_i(\sigma_i + \bar{q} b_{ii}) + \frac{1}{2} \sum_{k=1, k\neq i} \bar{q}\, \alpha_k e_k d_{ki} & i=j. \\ \frac{1}{2} \bar{q}(\alpha_i b_{ij} + \alpha_j b_{ji}) & i \neq j. \end{cases}$$
for some $\alpha_i > 0$.
(vi) ___
Then $x \equiv 0$ of (5.3) is uniformly asymptotically stable in the m.s.

Remark:
There is some mistake about the proof. But we can learn the idea and process.

Proof: Define $V(t,x) = \sum_{i=1}^{L} \alpha_i V^i(t, w_i)$.
$\Rightarrow$ (i) $\sum_{i=1}^{L} \alpha_i b_i(\|\psi^i(0)\|^2) \leq V(t, \psi(0)) \leq \sum_{i=1}^{L} \alpha_i a_i(\|\psi^i(0)\|^2)$.
For (ii) in Corollary 4.1. for $t \neq \tau_k$.
$\mathcal{L} V(t, x) = \sum_{i=1}^{L} \alpha_i \{ \mathcal{L}_i V^i(t, w_i) + g_i^T(t, x_t) V^i_{w_i}(t, w_i)$

$$+\frac{1}{2}\sum_{j=1,j\neq i}^{L} tr\left[\sigma_{ij}^T(t,W_t^j)\, V^i_{w^iw^i}(t,W^i)\, \sigma_{ij}(t,W_t^j)\right]\}$$

$$\le \sum_{i=1}^{L}\alpha_i\{\gamma_i c_i(V^i(t,W_i)) + c_i^{1/2}(\|W^i\|^2)\sum_{j=1}^{L}\bar{q}\, b_{ij}\, c_j^{1/2}(\|W^j\|^2)$$

$$+\frac{1}{2}\sum_{j=1,j\neq i}^{L}\bar{q}\, e_i\, \|\sigma_{ij}(t,W_t^j)\|^2\}$$

$$\le \sum_{i=1}^{L}\alpha_i\{\gamma_i c_i(V^i(t,W_i)) + c_i^{1/2}(\|W^i\|^2)\sum_{j=1}^{L}\bar{q}\, b_{ij}\, c_j^{1/2}(\|W^j\|^2)$$

$$+\frac{1}{2}\sum_{j=1,j\neq i}^{L}\bar{q}\, e_i\, d_{ij}\, c_j(\|W^j\|^2)\}$$

$$\triangleq z^T S z.$$

where $z^T = (c_1^{1/2}(\|W^1\|^2) \cdots c_L^{1/2}(\|W^L\|^2)) \in \mathbb{R}^L$.

S is defined in (V). Since S is negative-definite.

Set $\lambda_m(S)$ is the maximal eigenvalue of S. and $\lambda_m(S) < 0$.

$$\Rightarrow \mathcal{L}V(t,x) \le \underbrace{\lambda_m(S)}_{<0}\sum_{i=1}^{L} c_i(\|W^i\|^2). \qquad c_i = c_j,\ i\neq j.$$

$$= \lambda_m(S)\sum_{i=1}^{L} c(\|W^i\|^2) \qquad c \in K$$

$$\cancel{\le c(\sum \|W^i\|^2)}$$

For $t=\tau_k$, $V(\tau_k, x(\tau_k)) = \sum_{i=1}^{L}\alpha_i V^i(\tau_k, W^i(\tau_k))$

Since $V^i(\tau_k,\psi^i(0)) + J^i(\tau_k,\psi^i(\tau_k^-)) \le \hat{g}^i(V^i(\tau_k^-,\psi^i(0)))$.

$$\Rightarrow V(\tau_k,\psi(0)) + J(\tau_k,\psi(\tau_k^-)) = \sum_{i=1}^{L}\alpha_i V^i(\tau_k,\psi^i(0)) + J^i(\tau_k,\psi^i(\tau_k^-))$$

$$\le \sum_{i=1}^{L}\alpha_i\, \hat{g}^i(V^i(\tau_k^-,\psi^i(0)))$$

$$\triangleq \hat{g}(V(\tau_k^-,\psi(0))).$$

Thus all the conditions of Corollary 4.1 are satisfied. therefore $x=0$ of (5.3) is uniformly asymptotically stable. □

Example 5.1.

$$\begin{cases} dx = Ax\,dx + bf(y)\,dt + \sigma_{11}(x(t-1))\,dW_1 + \sigma_{12}(y)\,dW_2, & t\neq \tau_k \\ dy = (-\zeta y - \xi f(y))\,dt + a^T x\,dt + \sigma_{21}(x)\,dW_1 + \sigma_{22}(y(t-1))\,dW_2. & t\neq\tau_k \end{cases}$$

where $x^T = (x_1, \cdots, x_4)$ — system state. $y\in\mathbb{R}$ — controller. $A\in\mathbb{R}^{4\times4}$ $b\in\mathbb{R}^4$. $\zeta, \xi\in\mathbb{R}$. $f\in\mathbb{R}$ continuous $f(y)=0 \Leftrightarrow y=0$. and $0 < yf(y) < k|y|^2$. $\sigma_{11}\in\mathbb{R}^{4\times4}$, $\sigma_{21}\in\mathbb{R}^{4\times1}$. $\sigma_{21}\in\mathbb{R}^{4\times1}$. $\sigma_{22}\in\mathbb{R}^{1\times1}$. $W_1\in\mathbb{R}^4$. $W_2\in\mathbb{R}$.

$$\begin{cases} \Delta x(\tau_k) = J_1(\tau_k, x(\tau_k^-)) = \frac{1}{k^2}(-2x_1(\tau_k^-), -2x_2(\tau_k^-), 2x_3(\tau_k^-), 0)^T. \\ \Delta y(\tau_k) = J_2(\tau_k, y(\tau_k^-)) = -\frac{1}{1+k^2} y(\tau_k^-). \end{cases}$$

$$A = \begin{pmatrix} -5 & 0 & 0 & 0 \\ 0 & -6 & 0 & 0 \\ 0 & 0 & -8 & 0 \\ 0 & 0 & 0 & -10 \end{pmatrix}, \quad \sigma_{11} = 0.01 \begin{pmatrix} \sin x_1(t-1) & 0 & \frac{x_2(t-1)}{1+x_4^2} & 0 \\ 0 & \frac{x_2(t-1)}{1+x_1^2} & 0 & -x_3^2(t-1) \\ 0 & 0 & x_3(t-1) & 0 \\ 0 & 0 & 0 & -x_4(t-1) \end{pmatrix}.$$

$b^T = (1, 1, 1, 1)$. $a^T = (1, 1, 1, 1)$, $\zeta = 5$. $\xi = 2$. $\sigma_{12} = 0.01 \frac{y}{1+y^2}$.

$\sigma_{21}^T = 0.01(x_2, x_1, x_4, x_3)$. $\sigma_{22} = 0.01 \sin y(t-1)$.

Let $V^1(x) = \|x\|^2$. $V^2(y) = y^2$.

$$\begin{cases} dx = Ax\,dt + \sigma_{11}(x(t-1))\,dW_1 & S_1 \\ dy = (-\zeta y - \xi f(y))\,dt + \sigma_{22}(y(t-1))\,dW_2. & S_2 \end{cases}$$

$$\mathcal{L}_1 V^1(x) = 2x^T A x + \frac{1}{2} tr[\sigma_{11}^T(x(t-1)) \cdot 2 \cdot \sigma_{11}(x(t-1))]$$

$$\leq -10\|x\|^2 + 0.0001[\sin^2(x_1(t-1)) + \frac{x_2^2(t-1)}{(1+x_1^2)^2} + \frac{x_2^2(t-1)}{(1+x_4^2)^2} + x_3^2(t-1) + x_3^4(t-1) + x_4^2(t-1)].$$

$$\leq (-10 + 0.0002\bar{q}) \|x\|^2.$$

$\int_q^{2q} \frac{ds}{s} <$ $\quad c(s) = s \quad$ $\bar{q} = 2$.

$\ln s |_q^{2q}$

$$\mathcal{L}_2 V^2(y) = -2\zeta y^2 - 2\xi y f(y) + \frac{1}{2} \cdot 2 \cdot \sigma_{22}^2(y(t-1))$$

$$= -2\zeta y^2 - 2\xi \underbrace{y f(y)}_{>0} + 0.0001 \sin^2 y(t-1).$$

$\ln 2 < 0.4$.

$$\leq (-2\zeta + 0.0001\bar{q}) y^2 = (-10 + 0.0001\bar{q}) y^2.$$

$\sigma_1 = -10 + 0.0002\bar{q}$. $\quad \sigma_2 = -10 + 0.0001\bar{q}$. $\quad \bar{q} = 2$

$$V_x^{1\,T}(x)\, g_1(x, y) = 2x^T b f(y) \leq 4\|x\| |f(y)| \leq 4\|x\|\, k|y|$$

$(x_1 + x_2 + x_3 + x_4)^2 \leq 4(x_1^2 + \cdots + x_4^2) = 4\|x\|^2$.

$$V_y^{2\,T}(y)\, g_2(x, y) = 2y \cdot a^T x \leq 4\|x\| |y|.$$

$|f(y)| < k|y|$.

$\frac{1}{2} tr(\sigma_{12}^T(y) V_{xx}^1 \sigma_{12}(y)) = 0.00001\, y^2$. $\quad \frac{1}{2} tr(\sigma_{21}^T(x) V_{yy}^1(y) \sigma_{21}(x)) = 0.00001 \|x\|^2$.

$$\mathcal{L} V(x, y) = \alpha_1 [\mathcal{L}_1 V^1(x) + g_1^T(x,y) V_x^{1}(x)] + \alpha_2 [\mathcal{L}_2 V^2(y) + g_2^{(x,y)} V_y^2(y)]$$

$$+ \alpha_1 \tfrac{1}{2} tr[\sigma_{12}^T(y) V_{xx}^1(x) \sigma_{12}(y)] + \alpha_2 \tfrac{1}{2} tr[\sigma_{21}^T(x) V_{yy}^1(y) \sigma_{21}(x)]$$

$$\le \alpha_1 \sigma_1 \|x\|^2 + \underbrace{\alpha_1 4k \|x\| |y|} + \alpha_2 \sigma_2 |y|^2 + \underbrace{\alpha_2 4 \|x\| |y|}$$

$$+ \alpha_1\, 0.00001\, y^2 + \alpha_2\, 0.00001 \|x\|^2$$

$$= (x^T, y) \underbrace{\begin{pmatrix} \alpha_1\sigma_1 + \alpha_2 0.0001 & 2\alpha_1 k + 2\alpha_2 \\ 2\alpha_1 k + 2\alpha_2 & \alpha_2\sigma_2 + \alpha_1 0.0001 \end{pmatrix}}_{= S} \begin{pmatrix} x \\ y \end{pmatrix}$$

$x_i a_{ij} y_j$

Fix $\alpha_1 = 1$. $\alpha_2 = 1$.

$$S = \begin{pmatrix} -9.9995 & 2k+2 \\ 2k+2 & -9.9997 \end{pmatrix}$$ negative definite for $k < 3.9998$.

$\begin{pmatrix} -2 \\ -\frac{3}{2} \\ 1 \end{pmatrix} \begin{pmatrix} x \end{pmatrix}$

$\lambda(S) = -15.082$ $\lambda_2(S) = -2.917$ $k = 2$

$\frac{4}{k^2}$

$$V(x(\tau_k), y(\tau_k)) = \|x(\tau_k)\|^2 + y^2(\tau_k)$$

$$\le (1 + \tfrac{2}{k^2}) \|x(\tau_k^-)\|^2 + (1 - \tfrac{5}{1+k^2}) y^2(\tau_k^-)$$

$$\cancel{\le \alpha_M(d_k) V}$$

$$\le (1 + \tfrac{2}{k^2})(\|x\tau_k^-\|^2 + y^2(\tau_k^-))$$

$$= (1 + \tfrac{2}{k^2}) V(x(\tau_k^-), y(\tau_k^-))$$

Choosing $\mu = 0.04$. $k = 2$. For isolated system

$a_i(s) = b_i(s) = s^2$. $\longrightarrow$ ensure the asymptotic stability in the m.s. of isolated subsystems.

The eigenvalue of matrix $S$ are $-15.082$, $-2.917$.

Choose $\sigma = 2.917$ and $\mu > 0.14$. $\Rightarrow$ $(x, y)^T = (0, 0)$ is exponential stable in the m.s.